

Notes on the Theory of Average-Reward MDPs

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In this note, we study infinite-horizon Markov Decision Processes under the expected average-reward optimality criterion. We often refer to them as infinite-horizon average-reward MDPs, or simply, average-reward MDPs. Average-reward MDPs could be more interesting than discounted MDPs in that they model situations where future rewards are not discounted. Also, they may render more useful than finite-horizon MDPs as they could model some problems with a random number of interactions (e.g., in games). Solving average-reward MDPs in general exhibits more difficulty than discounted or finite-horizon MDPs. Yet, as we will see, for many such MDPs of interest, efficient procedures to compute an optimal policy exist. [1, 2] provide an elegant treatment of this subject. The results in this note are mostly adopted from [1, Chapters 8-9].

Notations. We introduce notations that will be used throughout. Given a set A , $\Delta(A)$ denotes the simplex of probability distributions over A . $\mathbb{I}\{A\}$ denotes the indicator function of event A , namely it equals 1 if A occurs, else 0. $\mathbb{N}$ and $\mathbb{R}$ denotes the set of naturals and reals, respectively. Unless specified, $\|\cdot\|$ is used to denote a generic norm. We write $X \sim p$ to indicate that X is drawn randomly from distribution p . Finally, given a set A , $|A|$ denotes the cardinality of A , and a function $f : A \rightarrow \mathbb{R}^{|A|}$ could interchangeably be viewed as a $|A|$ -dimensional vector.

1 Average-Reward MDPs: Definition

An average-reward MDP M is a tuple $M = (\mathcal{S}, \mathcal{A}, P, R)$, where:

- $\mathcal{S}$ denotes the state-space, which is the set of all possible states the agent may occupy;
- $\mathcal{A}$ denotes the action-space, which is the set of all possible actions available to the agent at any state;
- $P : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S})$ denotes the transition function such that $P(\cdot|s, a)$ is the probability distribution of the next-states when action $a \in \mathcal{A}$ in state $s \in \mathcal{S}$ is selected;
- $R : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathbb{R})$ denotes the reward function such that $R(s, a)$ denotes the probability distribution of (possibly random) rewards obtained when choosing action $a \in \mathcal{A}$ in state $s \in \mathcal{S}$.

The state space $\mathcal{S}$ and action space $\mathcal{A}$ maybe finite, countably infinite, or even continuous. Here, we restrict the attention to discrete $\mathcal{S}$ and $\mathcal{A}$ (i.e., finite or countably infinite). Then, for $s \in \mathcal{S}$ and $a \in \mathcal{A}$, we have $\sum_{x \in \mathcal{S}} P(x|s, a) = 1$. Furthermore, we assume that the reward and transition functions are stationary.

Remark 1 This definition can be readily extended to capture state-dependent action-spaces, namely where the set of actions differs in various states. In such cases, we write $\mathcal{A}_s$ to denote the set of actions available at state $s \in \mathcal{S}$. Unless stated otherwise, we assume $\mathcal{A}_s = \mathcal{A}$ for all $s \in \mathcal{S}$.

In the case of finite or countably infinite states (resp. actions), the state-space (resp. action-space) is generally termed **discrete**. An MDP is said to be discrete if both its state-space and action-space are discrete. An MDP is called finite if both its state-space and action-space are finite. For finite MDPs, we denote $S := |\mathcal{S}|$ and $A := |\mathcal{A}|$.

In this note, we restrict the attention to the case of finite MDPs, for which with no loss in generality we assume that $\mathcal{S} = \{1, \dots, S\}$ and $\mathcal{A} = \{1, \dots, A\}$ (unless stated otherwise). It is worth mentioning that some of the presented results hold in greater generality.

In a discrete MDP M , the transition function P is a collection of discrete probability distributions or probability vectors. $P(\cdot|s, a)$ denotes the probability distribution of next-states for $s \in \mathcal{S}$ and $a \in \mathcal{A}$. Hence, $\sum_{x \in \mathcal{S}} P(x|s, a) = 1$. We may also treat $P(\cdot|s, a)$ as a probability vector of dimension $S := |\mathcal{S}|$.

The interaction between MDP and the Agent. Consider a time slotted system, where time is divided into slots (or periods) of identical lengths indexed by $t \in \mathbb{N}$. The interaction between the agent and the MDP (environment) proceeds as follows. The agent starts in one initial state s_1 , which is determined by Nature, according to some distribution that could be unknown to the agent. At each time period $t \in \mathbb{N}$, the agent is in state s_t . At the beginning of time period t , she chooses an action $a_t \in \mathcal{A}_{s_t}$ according to some (possibly randomized) action selection rule and executes it. Upon executing a_t in s_t , the agent receives a (possibly random) reward r_t from the environment, which is sampled from the reward distribution $R(s_t, a_t)$, namely, $r_t \sim R(s_t, a_t)$. The agent may receive r_t immediately, or later during the period: in one shot or multiple phases. However, we do assume that she will earn the entire r_t until the end of the period and before deciding a_{t+1} . Upon executing a_t in s_t , the environment generates a next-state s_{t+1} according to $P(\cdot|s_t, a_t)$, i.e., $s_{t+1} \sim P(\cdot|s_t, a_t)$. Then, once the new slot begins, the environment transits to s_{t+1} and a new round of decision step starts. This process continues indefinitely. This interaction is depicted in Figure 1. It yields a history h_t at each time t :

$$h_t = (s_1, a_1, s_2, a_2, \dots, s_{t-1}, a_{t-1}, s_t).$$

The goal of the agent is to maximize the average (or per-step) expected reward collected during her interaction with the environment. More formally, she seeks to solve:

$$\max_{\text{strategies}} \lim_{N \rightarrow \infty} \frac{1}{N} \mathbb{E} \left[\sum_{t=1}^N r_t \right]$$

for any choice of initial state. Here, maximization is done over all possible decision making (action selection) strategies. Moreover, the expectation is taken with respect to possible randomness in the initial states, the randomness in the received rewards and state transitions, and possible randomization in the action selection rule. This defines a notion of optimality, which is called *the average expected reward optimality criterion* or for short *the average-reward optimality criterion*. The MDP M together with this criterion defines an **infinite-horizon average-reward Markov Decision Problem**.

Remark 2 *The definition above may suggest that average-reward MDPs, up to a normalization factor, could be related to discounted MDPs when the discount factor approaches 1. Similarly, they could be contrasted to finite-horizon MDPs, up to a normalization constant, when the horizon H tends to infinity. However, none of these observations proves instrumental to efficiently solve average-reward MDPs.*

Assumption 1 *In the rest of this note, we assume deterministic reward functions. That is, we assume choosing a in s yields $r = r(s, a) = R(s, a)$. We further assume that (deterministic) rewards are uniformly bounded, namely there exists a constant $R_{\max} < \infty$ such that*

$$\sup_{s \in \mathcal{S}} \sup_{a \in \mathcal{A}} |R(s, a)| < R_{\max}.$$

As in previous MDP settings, we remark that the presented results below will hold for the case of stochastic rewards under very mild assumptions. In the case of stochastic rewards, it suffices to replace $R(s, a)$ in them with $\mu(s, a) := \mathbb{E}_{r \sim R(s, a)}[r]$.

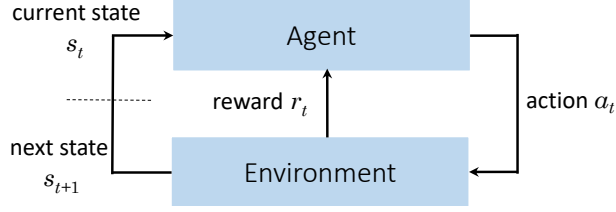


Figure 1: The interaction between the agent and the environment (MDP), where $s_{t+1} \sim P(\cdot | s_t, a_t)$ and $r_t \sim R(s_t, a_t)$.

2 Gain and Bias Functions

In this section, we introduce two important functions that play the role of value function in discounted MDPs and finite-horizon MDPs. The first one is the average-reward function (or the gain function) of a policy, and the second is the bias function of a policy. *The notions of policy in average-reward MDPs and discounted MDPs are identical. We thus omit their definition here and refer to the notes on discounted MDPs.*

Let $\pi \in \Pi^{\text{HR}}$ in M . The **gain or (average-reward)** of π is a function $g_M^\pi : \mathcal{S} \rightarrow \mathbb{R}$ defined as

$$g_M^\pi(s) := \lim_{N \rightarrow \infty} \frac{1}{N} \mathbb{E}^\pi \left[\sum_{t=1}^N r(s_t, a_t) \middle| s_1 = s \right],$$

where the expectation $\mathbb{E}^\pi$ is taken over trajectories generated by π . Here, the subscript M indicates that the underlying MDP is M . In fact, $g_M^\pi(s)$ represents the average-reward per step obtained in a steady state regime when the system starts in s and always follows π . For brevity, we denote $g^\pi := g_M^\pi$ whenever it is clear from the context that the underlying MDP is M .

In the case of bounded rewards, the above limit exists and also interchanging the expectation and the limit is allowed. Further, under this assumption, we have $g^\pi(s) \leq R_{\max}$.

Using the notion of gain, we can rewrite the optimization in average-reward MDPs as follows:

$$g^* = \max_{\pi \in \Pi^{\text{HR}}} g^\pi$$

The quantity g^* is called **the optimal gain**, and any policy achieving g^* is said to be (gain-)optimal (in the underlying MDP M). Optimization over the class of history-dependent randomized policies is notoriously difficult. However, the following results allow us to restrict attention to the much simpler class of stationary deterministic policies.

Theorem 1 *Assume that M is discrete. Then an optimal stationary deterministic policy exists.*

This theorem tells us that the optimal gain satisfies:

$$g^* = \max_{\pi \in \Pi^{\text{SD}}} g^\pi$$

Therefore, in what follows, we provide results only for stationary (randomized) policies, since they will be the class of policies of interest.

Next we introduce the bias functions. For a policy $\pi \in \Pi^{\text{SR}}$ in M , **the bias function** of π is a function $b_M^\pi : \mathcal{S} \rightarrow \mathbb{R}$ and is defined as

$$b_M^\pi(s) = \mathbb{E}^\pi \left[\sum_{t=1}^{\infty} \left(r(s_t, a_t) - g_M^\pi(s_1) \right) \middle| s_1 = s \right].$$

For brevity, we denote $b^\pi := b_M^\pi$ whenever there is no ambiguity about the underlying MDP.

The bias can be interpreted as follows. The expected N -step cumulative reward under π and starting in s is $\mathbb{E}^\pi \left[\sum_{t=1}^N r(s_t, a_t) \middle| s_1 = s \right]$ but in the steady state $Ng^\pi(s)$ units of rewards will be collected in N steps. In general, $\mathbb{E}^\pi \left[\sum_{t=1}^N r(s_t, a_t) \middle| s_1 = s \right] \neq Ng^\pi(s)$ because of *the transient* time steps. So we might be interested in quantifying how much rewards is lost/gained due to this transient regime. The bias $b^\pi(s)$ indeed measures the difference between the two quantities (in the limit when $N \rightarrow \infty$) and reflects the behavior of the transient regime in the system (in terms of collected rewards).

Now suppose that M and π are such that $g^\pi(s) = g^\pi$ for all $s \in \mathcal{S}$, i.e., the gain of π in M does not depend on the starting state —this is the case in, e.g., ergodic MDPs, which will be discussed later. In such a case, we have

$$\begin{aligned} b^\pi(s) - b^\pi(s') &= \mathbb{E}^\pi \left[\sum_{t=1}^{\infty} r(s_t, a_t) \middle| s_1 = s \right] - g^\pi - \mathbb{E}^\pi \left[\sum_{t=1}^{\infty} r(s_t, a_t) \middle| s_1 = s' \right] + g^\pi \\ &= \mathbb{E}^\pi \left[\sum_{t=1}^{\infty} r(s_t, a_t) \middle| s_1 = s \right] - \mathbb{E}^\pi \left[\sum_{t=1}^{\infty} r(s_t, a_t) \middle| s_1 = s' \right]. \end{aligned}$$

In other words, in this case $b^\pi(s) - b^\pi(s')$ measures the amount of rewards the agent *wins* (in expectation) when she starts in s instead of s' .

2.1 Classification of MDPs

Based on the reachability of various states from each other, we can classify MDPs as follows.

1. An MDP is said to be **ergodic** or **recurrent** if every state is reachable from any other state under every stationary deterministic policy. Formally,

$$\forall s_1, s_2 \in \mathcal{S}, \forall \pi \in \Pi^{\text{SD}}, \exists t < \infty : (P^\pi)^t(s_2, s_1) > 0.$$

Alternatively, the transition matrix of the Markov chain induced by every stationary deterministic policy has a single recurrent class.

2. An MDP is said to be **unichain** if the transition matrix of the Markov chain induced by every stationary deterministic is unichain, namely it has a single recurrent class plus a set of transient states. If the transient set is empty, then it coincides with ergodic MDPs.
3. An MDP is said to be **communicating** if every state is reachable from any other state under some stationary deterministic policy. Formally,

$$\forall s_1, s_2 \in \mathcal{S}, \exists \pi \in \Pi^{\text{SD}}, \exists t < \infty : (P^\pi)^t(s_2, s_1) > 0.$$

4. An MDP is said to be **weakly communicating** if the state-space can be partitioned into two subsets as follows: A closed set of states in which every state is reachable from every other state in the set under some stationary deterministic policy, and a set of states that are transient under all policies.
5. An MDP is called **multichain** if there exists some stationary policy whose induced Markov chain has two or more irreducible recurrent classes.

A weakly communicating MDP can be seen as a communicating MDP with *extra transient states*. Similarly, a unichain MDP can be seen as an ergodic MDP with extra transient states. The above definitions reveal the following hierarchy:

- If M is ergodic, then M is unichain too but the converse may not be true.
- If M is ergodic, then M is communicating too but the converse may not be true.
- If M is communicating, then M is weakly communicating too but the converse may not be true.
- If M is unichain, then M is weakly communicating too but the converse may not be true.

Examples. Consider the RiverSwim MDP shown in Figure 2. Clearly it is communicating since every state is reachable from any other state. However it is not ergodic since, for example, state 5 is not reachable from state 4 by a policy always prescribing the **left** action. On the other hand, one can easily verify that the MDP shown in Figure 3 is ergodic.

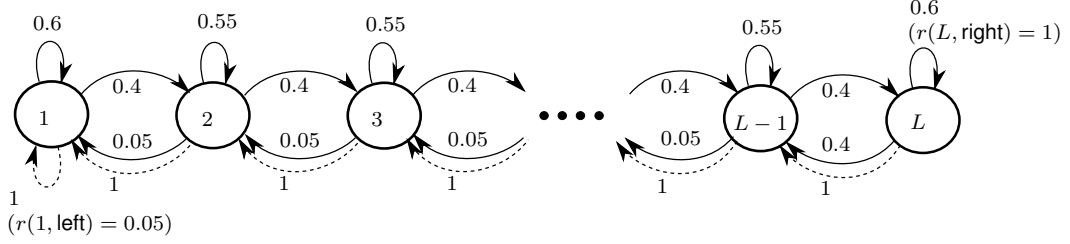


Figure 2: The RiverSwim MDP [3]

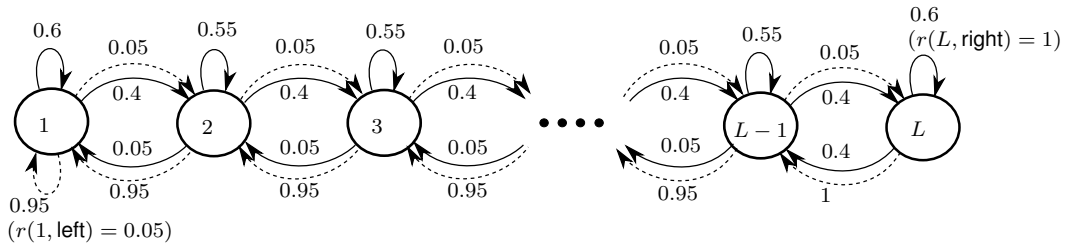


Figure 3: The Ergodic RiverSwim MDP

In unichain MDPs, the gain of any stationary policy does not depend on the starting state, namely it is a constant. Formally, $g^\pi(s) = g^\pi$ for all $s \in \mathcal{S}$ and all $\pi \in \Pi^{\text{SR}}$. In weakly communicating MDPs, the gain of a stationary policy may depend on the starting state. However, the optimal gain does not depend on the starting state, i.e., $g^*(s) = g^*$ for all $s \in \mathcal{S}$.

In relation to this classification, we introduce the notion of **diameter** of the MDP as defined in [4]:

Definition 1 (Diameter [4]) The diameter D of an MDP M is $D = \max_{s,s' \in \mathcal{S}, s \neq s'} \min_{\pi \in \Pi^{\text{SD}}} \mathbb{E}[T^\pi(s, s')]$, where $T^\pi(s, s')$ denotes the number of steps it takes to get to s' starting from s and following policy $\pi \in \Pi^{\text{SD}}$.

By definition, any communicating MDP has a finite diameter. More precisely, an MDP is communicating if and only if it has a finite diameter. Note that a unichain or weakly communicating MDP may not have a finite diameter (why?).

2.2 Evaluating Gain and Bias

Let us remark that in average-reward MDPs, induced Markov reward processes (MRPs) by stationary policies are defined similarly to the case of discounted MDPs. Recall that for a stationary policy $\pi \in \Pi^{\text{SR}}$, P^π and r^π used to denote the transition matrix and reward vector of the MRP induced by π , respectively. Specifically, for $s, s' \in \mathcal{S}$,

$$P_{s,s'}^\pi = \sum_{a \in \mathcal{A}} \pi(a|s) P(s'|s, a), \quad r^\pi(s) = \sum_{a \in \mathcal{A}} \pi(a|s) R(s, a).$$

We will derive closed-form formulas for g^π and b^π for stationary policies. To this end, let us first write the N -step accumulated reward in the MRP induced by π . For any $N \in \mathbb{N}$,

$$\begin{aligned} \mathbb{E}^\pi \left[\sum_{t=1}^N r(s_t, a_t) \middle| s_1 = s \right] &= \mathbb{E}^\pi[r(s_1, a_1)|s_1 = s] + \mathbb{E}^\pi[r(s_2, a_2)|s_1 = s] + \dots + \mathbb{E}^\pi[r(s_N, a_N)|s_1 = s] \\ &= r^\pi(s) + [P^\pi r^\pi](s) + \dots + [(P^\pi)^{N-1} r^\pi](s) = \sum_{t=1}^N [(P^\pi)^{t-1} r^\pi](s) \end{aligned}$$

where we have used that for any $t \geq 1$, when following $\pi \in \Pi^{SR}$, $\mathbb{P}(s_t = y | s_1 = x) = (P^\pi)^t(x, y)$.

Definition 2 For a row stochastic matrix A , **the limiting matrix** or **the Cesaro-average** of A is defined as

$$\bar{A} := \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{t=1}^N A^{t-1}$$

with the convention $A^0 = I$, where I is the identity matrix.

Note that when $\mathcal{S}$ is discrete, the Cesaro-average in Definition 2 exists. Hence, for discrete MDPs, we can write the gain of a stationary π in terms of the Cesaro-average of P^π as follows

$$g^\pi = \bar{P}^\pi r^\pi$$

Thus, we have shown:

Theorem 2 Let $\pi \in \Pi^{SR}$. Then, $g^\pi = \bar{P}^\pi r^\pi$.

We now consider the case when π induces an MRP, which is unichain or irreducible. In this case, the gain is a constant function, namely, it is independent of the starting state. We have:

Theorem 3 (Bellman Equation for Policy π) Let $\pi \in \Pi^{SR}$. Assume that π induces an MRP, which is irreducible or unichain. Then

$$g^\pi \mathbf{1} = \bar{P}^\pi r^\pi.$$

Furthermore, the bias function b^π satisfies the Bellman equation:

$$g^\pi \mathbf{1} + (I - P^\pi)b^\pi = r^\pi.$$

As a result,

$$b^\pi = (I - P^\pi + \bar{P}^\pi)^{-1}(I - \bar{P}^\pi)r^\pi + c\mathbf{1},$$

where $c \in \mathbb{R}$ is arbitrary.

Note that the matrix $I - P^\pi + \bar{P}^\pi$ is non-singular, so the last assertion in Theorem 3 is well-defined.

3 Finding an Optimal Policy

3.1 Bellman's Optimality Equations

As in discounted and finite-horizon MDPs, in average-reward MDPs a variant of Bellman's optimality equations could be instrumental to characterize an optimal policy in Π^{SD} . More precisely, in this case, the Bellman's optimality equations characterize the optimal gain and bias functions, as formalized in the following theorem:

Theorem 4 The Bellman optimality equations for an average-reward MDP are given by:

$$g^*(s) + b^*(s) = \max_{a \in \mathcal{A}} \left(R(s, a) + \sum_{x \in \mathcal{S}} P(x|s, a)b^*(x) \right), \quad \forall s \in \mathcal{S}. \quad (1)$$

In particular, if M is weakly communicating, then:

$$g^* + b^*(s) = \max_{a \in \mathcal{A}} \left(R(s, a) + \sum_{x \in \mathcal{S}} P(x|s, a)b^*(x) \right), \quad \forall s \in \mathcal{S}. \quad (2)$$

Furthermore, $\pi \in \Pi^{SD}$ is optimal if and only if it attains the maximum in the Bellman optimality equations: For all $s \in \mathcal{S}$,

$$\pi(s) \in \operatorname{argmax}_{a \in \mathcal{A}} \left(R(s, a) + \sum_{x \in \mathcal{S}} P(x|s, a)b^*(x) \right).$$

In the optimality equations (2), g^* is uniquely defined. However, the optimal bias function b^* is defined up to an additive constant. In fact, it can be verified that if b^* is a solution to (2), so is $b^* + c\mathbf{1}$ for any $c \in \mathbb{R}$.

3.2 Value Iteration for Average-Reward MDPs

Similarly to the case of discounted MDPs, we can devise a variant of value iteration (VI) to solve the Bellman's optimality equations (2), or those for the general case (1). Here, we only discuss the case of weakly communicating MDPs.

The algorithm is almost identical to VI used for discounted MDPs, except that it does not use any discount factor and that it has a slightly different stopping criterion. Precisely speaking, VI for average-reward MDPs proceeds as follows. It takes as input a parameter $\varepsilon > 0$, and initializes vector V_0 , which can for instance be set to $\mathbf{0}$. At each iterate n , VI maintains an approximate V_n of the optimal value function V^* , which is updated as follows:

$$V_{n+1}(s) = \max_{a \in \mathcal{A}} \left(R(s, a) + \sum_{x \in \mathcal{S}} P(x|s, a) V_n(x) \right), \quad s \in \mathcal{S}.$$

$g_n = V_{n+1} - V_n$ will serve as an approximation of g^* . This is repeated until

$$\max_{s \in \mathcal{S}} (V_{n+1}(s) - V_n(s)) - \min_{s \in \mathcal{S}} (V_{n+1}(s) - V_n(s)) < \varepsilon.$$

Finally, it outputs the greedy policy w.r.t. V_n , namely, any policy that maximizes $R(s, a) + \sum_{x \in \mathcal{S}} P(x|s, a) V_{n+1}(x)$ over actions $a \in \mathcal{A}$. The pseudocode of VI for average-reward MDPs is presented in Algorithm ??.

Algorithm 1 Value Iteration (VI) for Average-Reward MDPs

Input: ε
 Select $V_0 \in \mathbb{R}^{\mathcal{S}}$ arbitrarily
 $n = 0$
repeat
 $n = n + 1$
 Update, for each $s \in \mathcal{S}$,

$$V_n(s) = \max_{a \in \mathcal{A}} \left(r(s, a) + \sum_{x \in \mathcal{S}} P(x|s, a) V_{n-1}(x) \right)$$

until $\max_{s \in \mathcal{S}} (V_n(s) - V_{n-1}(s)) - \min_{s \in \mathcal{S}} (V_n(s) - V_{n-1}(s)) < \varepsilon$

Output:

$$\pi^{\text{VI}}(s) = \operatorname{argmax}_{a \in \mathcal{A}} \left(r(s, a) + \sum_{x \in \mathcal{S}} P(x|s, a) V_n(x) \right), \quad s \in \mathcal{S}$$

VI can be shown to converge in general MDPs as long as Markov chains induced by all stationary deterministic policies are aperiodic.¹

Note that the update in the VI algorithm above can be concisely written as $V_{n+1} = \mathcal{T}V_n$, where $\mathcal{T} : \mathbb{R}^{\mathcal{S}} \rightarrow \mathbb{R}^{\mathcal{S}}$ is an operator defined as follows: For all $f : \mathcal{S} \rightarrow \mathbb{R}$,

$$(\mathcal{T}f)(s) := \max_{a \in \mathcal{A}} \left(R(s, a) + \sum_{x \in \mathcal{S}} P(x|s, a) f(x) \right).$$

The sequence $(V_n)_{n \geq 1}$ may diverge, but the sequence $(Z_n)_{n \geq 1}$, with Z_n defined as

$$Z_n = \max_{s \in \mathcal{S}} (V_{n+1}(s) - V_n(s)) - \min_{s \in \mathcal{S}} (V_{n+1}(s) - V_n(s)),$$

¹In the absence of this condition, one apply the so-called *aperiodicity transformation*, which perturbs the transition of the MDP in a way to ensure that the induced chains are aperiodic; see, [1, Chapter 9].

may converge. It can be shown that $Z_{n+1} \leq \kappa Z_n$, where

$$\kappa := \max_{s,a} \max_{s',a'} \left(1 - \sum_{j \in \mathcal{S}} \min\{P(j|s, a), P(j|s', a')\} \right).$$

Now, if $\kappa < 1$, the stopping criterion of VI will be satisfied for any ε after *finitely many* iterations. Note that convergence of VI will also happen even if $\kappa = 1$; see [1, Proposition 8.5.1].

The following theorem summarizes the convergence guarantees of VI :

Theorem 5 ([1, Theorem 9.4.5]) *Suppose all stationary deterministic optimal policies induce aperiodic Markov chains. Then, for any $V_0 \in \mathbb{R}^{\mathcal{S}}$,*

$$\lim_{n \rightarrow \infty} (V_{n+1} - V_n) = g^*.$$

Furthermore,

$$\min_{x \in \mathcal{S}} (V_{n+1}(x) - V_n(x)) \leq g^{\pi_n}(s) \leq g^* \leq \max_{x \in \mathcal{S}} (V_{n+1}(x) - V_n(x))$$

where π_n is the greedy policy derived using V_n , i.e.,

$$\pi_n(s) \in \operatorname{argmax}_{a \in \mathcal{A}} \left(R(s, a) + \sum_{x \in \mathcal{S}} P(x|s, a) V_n(x) \right), \quad s \in \mathcal{S}.$$

Hence, π_n is ε -optimal.

In particular, for weakly communicating MDPs, in which $g^*(s) = g^*$ for all s , we have, $\lim_{n \rightarrow \infty} (V_{n+1} - V_n) = g^*$ and that

$$\max_{s \in \mathcal{S}} (V_{n+1}(s) - V_n(s)) - \min_{s \in \mathcal{S}} (V_{n+1}(s) - V_n(s)) \leq \max_{s \in \mathcal{S}} (V_n(s) - V_{n-1}(s)) - \min_{s \in \mathcal{S}} (V_n(s) - V_{n-1}(s)),$$

so that the *span* of $V_{n+1} - V_n$ shrinks monotonically.

Finally, note that the following can be used as an estimate of g^* :

$$\frac{1}{2} \left[\max_{s \in \mathcal{S}} (V_{n+1}(s) - V_n(s)) + \min_{s \in \mathcal{S}} (V_{n+1}(s) - V_n(s)) \right]$$

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